MIL-O-23560/4A(Wep)
25 August 1964
Superseding
MIL-O-23560/4(Wep)
10 January 1963

MILITARY SPECIFICATION

OSCILLOGRAPH, TYPE IV

(For Torpedo Mark 37 Mod 1)

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 Scope. This specification covers the detail requirements for the design and procurement of Oscillograph, Type IV, BUWEPS Drawing 2436248.

2. APPLICABLE DOCUMENTS

2.1 The following specifications of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification.

SPECIFICATIONS

MILITARY

MIL-O-23560(Wep)	Oscillographs, General Specification for

(Copies of specifications required by contractors in conjunction with specific procurement functions should be ordered from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Requirements shall be in accordance with Specification MIL-O-23560 (Wep) and as specified herein.

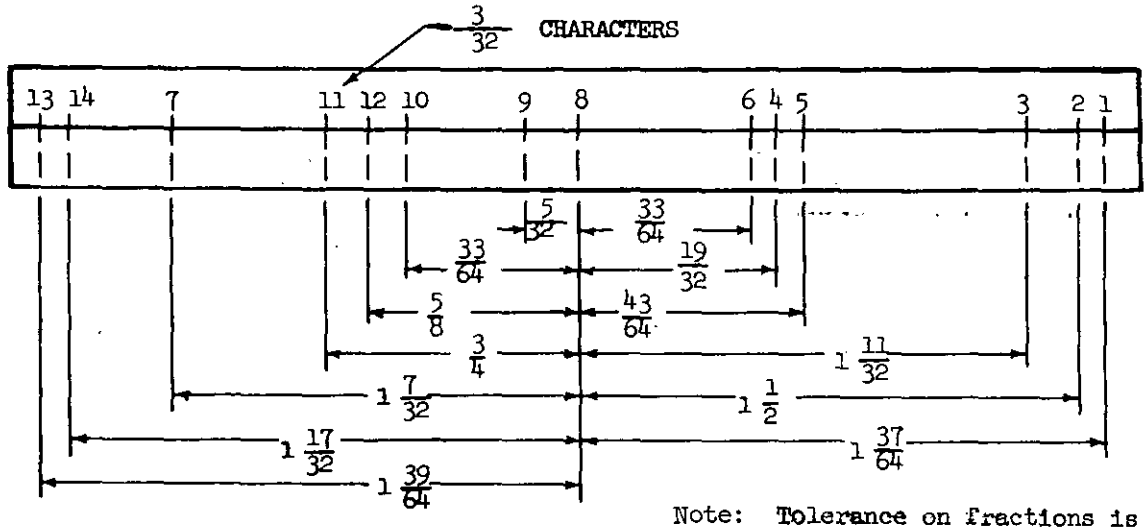
3.2 Design and construction shall be in accordance with BUWEPS Drawing 2436248 and Specification MIL-O-23560(Wep).

3.2.1 The viewing screen shall be marked as shown in Figure 1. The horizontal line shall coincide with the vertical centerline of the film magazine exposure slot. Position 8 shall coincide with the horizontal centerline of the exposure slot.

FSC 1355

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3.2.2 The galvanometers shall be adjusted so that the neutral position (no signal applied) of the traces shall fall on the corresponding marks of the viewing screen. The trace of channel 1 shall be at the top of the film, when the film is viewed emulsion side up and the starting end is at the observer's left.



Note: Tolerance on fractions is ±1/64

Figure 1

3.2.3 The galvanometers shall be in accordance with Specification MIL-O-23560(Wep), and the galvanometer channels shall meet the requirements of Tables I and II.

3.2.4 Galvanometer channel 2 shall meet the damping requirements of Specification MIL-O-23560(Wep) when the resistive driving impedance of the driving source is 360 ohms plus or minus 5 percent.

3.2.5 Motor and lamp circuits. The motor and lamp circuits shall be in accordance with BUWEPS Drawing 2436248 and shall operate at 26 volts dc plus or minus 10 percent.

3.2.6 The oscillograph shall not require more than 2.5 amperes of dc current when the operating voltage of 3.2.5 is applied.

3.2.7 Weight. The weight of the oscillograph shall not exceed 10 pounds.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Testing and inspection shall be as described in Specification MIL-O-23560(Wep), except that the proper impedance-matching resistors shall be used to meet the requirements of 3.2.4 of Specification MIL-O-23560/4(Wep).

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with Specification MIL-O-23560(Wep).

6. NOTES

6.1 Intended use. The oscillograph is intended for use in the Torpedo Mk 37 Mod 1 (Exercise Head Mk 64 Mod 2).

6.2 The notes specified in Specification MIL-O-23560(Wep) are applicable.

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Channel Number	Pin Connection on P71 Polarity		Sensitivity		Direction of Deflection*
	Positive	Negative	Input to P71 Pins (ma)	Deflection at Viewing Screen (Inches)	
1	1	2	1.74 ma ± 2%	1.15 ± 20%	Down
2	3	4	0.035 ± 2%	2 ± 20%	Down
3	5	6	1.74 ma ± 2%	1.15 ± 20%	Down
4	7	8	"	"	Down
5	9	10	"	"	Down
6	11	12	"	"	Down
7	14	13	"	"	Up
8	15	16	"	"	Down
9	18	17	0.035 ± 2%	1.8 ± 20%	Up
10	20	19	1.74 ma ± 2%	1.15 ± 20%	Up
11	22	21	"	"	Up
12	24	23	"	"	Up
13	26	25	"	"	Up
14	--	--	Dummy	--	----

*When film is read emulsion side up and the starting end of film is at the observer's left.

Table I

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Channel Number	Pin Connections on P71		Natural Freq. (cps)	Voltage (rms) millivolts	Frequency Range (cps)	Frequency Response Flat Within		
						@ 0°F	@ 100°F	@ 130°F
1	1	2	500	3.5 ± 0.3	10 to 300	+ 7.5% - 15%	+ 7.5%	+ 15% -7.5%
2	3	4	100	0.6 ± 0.06	10 to 30	± 7.5%	± 7.5%	± 7.5%
3	5	6	500	3.5 ± 0.3	10 to 300	+ 7.5% - 15%	"	+15% -7.5%
4	7	8	"	"	"	"	"	"
5	9	10	"	"	"	"	"	"
6	11	12	"	"	"	"	"	"
7	13	14	"	"	"	"	"	"
8	15	16	"	"	"	"	"	"
9	17	18	100	0.6 ± 0.06	10 to 30	± 7.5%	"	±7.5%
10	19	20	500	3.5 ± 0.3	10 to 300	+ 7.5% - 15%	"	+15% -7.5%
11	21	22	500	3.5 ± 0.3	10 to 300	"	"	"
12	23	24	"	"	"	"	"	"
13	25	26	"	"	"	"	"	"
14	--	--	--	Dummy	--	--	--	--

Table II

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-O-23560/4A(Wep) Oscillograph, Type IV

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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Bureau of Naval Weapons
Washington, D. C. 20360

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